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Assignment: 7

Planning Process

My planning process for this assignment followed an iterative development approach using an AI assistant (Gemini):

1. **Analysis & Porting:** I started by providing the AI with the full assignment instructions (Ex7) and the reference C code from Ex3. The goal was to translate the logic to Python while implementing the new dynamic board rules (Connect-N vs. Tic-Tac-Toe).
2. **Logic Refinement:** I tested the initial code and identified logical discrepancies in the "Tic-Tac-Toe" mode (specifically, the requirement for Human vs. Human only). I prompted the AI to enforce this rule.
3. **Formatting:** I compared the AI's output with the specific execution examples provided in the assignment. I provided the AI with "Expected vs. Actual" examples to fix the board printing style (dots vs. numbers), headers, and menu options (adding the 'Random' and 'Strategic' bots).
4. **Code Quality & Style:** Finally, I refined the code to meet strict coding standards. This involved fixing input cursor placement (same line) and replacing all "Magic Numbers" with named constants.
5. **Final Adjustment (Forum Clarification):** Based on the new clarification in the course forum ('Same as Ex3... regarding 2 and 3 needs to be Tic-Tac-Toe'), I reverted the player selection menu to strictly 'human' vs 'computer' (removing the random/strategic split). The computer logic now defaults to the Ex3 heuristic algorithm. I also ensured that dimensions of 3 trigger the Tic-Tac-Toe mode.
6. **Fixing Logic based on Auto grader Feedback.**
7. **Fix for Auto grader Newline Issue fix the code according to this.**
8. **Final Separation of Board and Prompts: Fix the code:** The grader output shows '2 line errors'. The 'Your Output' showed the board glued to the player selection prompt (e.g., ...strategic computer: |.|.|.|). To fix this, I added a single print() call in main() immediately after the player selection phase completes, ensuring the board starts on a fresh line while keeping the selection prompts logically grouped.

Prompts Used

Step 1: Initial Implementation

תפתור את מטלה 7 שאלה הם הוראותיה שים לב לא לפספס דבר ולבדוק את עצמך:

EX7 2026 - Python

General instructions

The T.A responsible for this exercise is Shalom(Read more about the author).

You must fork the repository ex7 from the course's git.

Change ex7.py according to instructions.

After completing all the requirements, submit it to the submit system under ex7 till 28.1.2026 at 23:59:59.

Your submission should be a zip file containing:

.git folder (after fork and clone, as you did in ex0).

ex7.py

prompt.pdf - save your prompts to a pdf file

Don't miss this!!!

README.md (optional)

Late submissions will be fined accordingly:

One day - 10 points. (the maximum grade is 90)

Two days - 20 points.

Three days - 30 points.

You can't submit this or any other exercise more than 3 days late.

Please post any questions about the exercise publicly in the course forum.

Requests for an extension (for justified reasons only) should be sent to the email: CSI.BIU2024@gmail.com with:

Full Name.

Your Submit system username.

ID.

The input and output design instructions must be read carefully, exactly according to the attached examples.

There are executable programs (Linux) in the forum.

You can run them on your computer and it may help you understand better.

The automation grading system checks that your output is the same as the expected output and therefore your output must be exactly the same as the examples.

As so, there is a Coding Style file in the forum - read it and work accordingly.

Testing your code on the server is highly recommended to ensure there are no warnings or errors.

An exercise that will have warning(s) or error(s) will get a fine that can reach 100 points.

You are reminded that thinking together is a blessing, working together is cheating.

An exercise that the system will catch as a “copy” will result in 0 - for everyone involved, and additional actions will take place.

In the development of your code, you can use any work environment.

The main thing is that you know how to take the code files from this environment, check them on the university servers, and submit them using the submission system.

Please note that you submit only the files containing your code, and not unnecessary files created by the work environment.

Also, DO NOT submit files/folders with names containing Hebrew characters.

Please note that it is not possible to copy a file with a path containing Hebrew characters to the university's servers.

Exercise 7 - Upgraded Connect- 4 (Connect-N)

We all remember the good old days when the sun was shining, and trees' roots weren't upside down. This brings us back to simpler times when our great concern was ex3.

Your task is simple - recreate ex3, but in python using any AI tool you want (prompts you used will be submitted in a separate pdf file, did I mention that?), with a few key changes:

The function for printing the board is not given.

The board's dimensions are received from the user as inputs and need to be validated, and the game's goal will be adjusted accordingly by the logic explained later.

N-Connect Logic:

Since we want to make our board's size adjustable, we are going to change the length of the sequence we need to mark.

Anything less than 2 or greater than 100 (101 and above, rows or columns) - invalid.

If 2 is chosen for either rows or columns, sequence is 2 (very short game, but that is the user's responsibility)

If either is 3 - game transforms to tic-tac-toe and the board is 3X3 (Ix Igul), two human players only. You can assume no one is going to mark a taken cell.

4 <= rows or columns <= 5 sequence is 3

6 <= rows or columns <= 10 sequence is 4

11 and above - sequence is 5

Tic-Tac-Toe:

Users will need to enter 1 number for the cell chosen. Cells are numbered in this format:

|1|2|3|

|4|5|6|

|7|8|9|

Columns will not be numbered

Good Luck!

All other examples are the same as ex3

Prompt.pdf:

Computer science students are well known for their exquisite verbal skills, as well as social skills, so as a part of this assignment, you will attach a file that:

Describes your planning process in a few sentences. (for example: step1: xyz, step 2: xyz, but with chatGPT etc).

Shows the prompts you used in order to execute your mastermind plan

Note: I am neither limiting your creative integrity nor caging your artistic ambitions in this part. You can use any format you desire, as long as you make sure your work is understood. m

מטלה מספר 3 המוזכרת בה הייתה זו:

EX3 2026 - Functions and Arrays

General instructions

The T.A responsible for this exercise is Eyal Dayan.

You must fork the repository ex3 from the course's git.

Change ex3.c according to instructions.

After completing all the requirements, submit it to the submit system under ex3 til 10.12.25 at 23:59:59.

Your submission should be a zip file containing:

.git folder (after fork and clone, as you did in ex0).

ex3.c

README.md (optional)

Late submissions will be fined accordingly:

One day - 10 points. (the maximum grade is 90)

Two days - 20 points.

Three days - 30 points.

You can’t submit this or any other exercise more than 3 days late.

Please post any questions about the exercise publicly in the course forum.

Requests for an extension (for justified reasons only) should be sent to the email: CSI.BIU2024@gmail.com with:

Full Name.

Your Submit system username.

ID.

The input and output design instructions must be read carefully, exactly according to the attached examples.

There are executable programs (Linux) in the forum, each with different dimensions.

You can run them on your computer and it may help you understand better.

The automation grading system checks that your output is the same as the expected output and therefore your output must be exactly the same as the examples.

As so, there is a Coding Style file in the forum - read it and work accordingly.

Testing your code on the server is highly recommended to ensure there are no warnings or errors.

An exercise that will have warning(s) or error(s) will get a fine that can reach 100 points.

You are reminded that thinking together is a blessing, working together is cheating.

An exercise that the system will catch as a “copy” will result in 0 - for everyone involved, and additional actions will take place.

In the development of your code, you can use any work environment.

The main thing is that you know how to take the code files from this environment, check them on the university servers, and submit them using the submission system.

Please note that you submit only the files containing your code, and not unnecessary files created by the work environment.

Also, DO NOT submit files/folders with names containing Hebrew characters.

Please note that it is not possible to copy a file with a path containing Hebrew characters to the university's servers.

This exercise compile command (Linux) is:

```
gcc -std=c99 -Wall -Wextra -Werror -DNDEBUG -DROWS=6 -DCOLS=7
ex3.c -o ex3.out -lm
```

Please note the following orders about the compile command in the first section of the file.

Exercise 3 - Connect 4

In this exercise we will implement a version of the game Connect Four.

In the classic version of the game two players play against each other.

Each player has tokens of one shape and on each turn the player drops a token into a board that consists of columns and rows.

The goal of the game is to connect n tokens of the same color in a continuous vertical, horizontal or diagonal line before the opponent manages to do so.

In our version we will implement two types of players:

Human player - the program will ask the user to choose a column into which they want to insert a token.

Computer player - the computer will choose a column according to a priority order that will be defined later.

Game values via compilation flags

The board dimensions will be received as constant values through the compilation command using the -D flag.

So, if you want to test different dimensions you need to compile it with different flags.

for example, this compilation command:

gcc -std=c99 -Wall -Wextra -Werror -DNDEBUG -DROWS=8 -DCOLS=10 ex3.c -o ex3.out -lm

Will set the number of rows to 8, the number of columns to 10.

The code must work even if we change the board dimensions by changing these constants.

”Magic numbers” should not appear in the code.

You can assume the following assumption:

The minimum board dimension will be 1 [including 1].

Please note: you cannot change the name of those define values, and you cannot remove the #ifndef and the #endif before and after each defined value.

Program flow

After the compilation, we are running the program normally ./ex3.out

And then, we start:

We are printing the dimension and the sequence needed for the current game:

Connect Four (6 rows x 7 cols)

You are given this printf command in the git code.

Then, the user should be asked to choose the type of the first player and then the type of the second player. Any combination of players is possible (4 ordered options in total).

Choose type for player 1: h - human, c - computer: h

Choose type for player 2: h - human, c - computer: c

Please note, you cannot assume anything about the input, beside it will be ASCII characters.

Lucky for you, this function is also given to you in the git file.

After we get the players, we will:

1. Print an empty game board.

```
|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

 1 2 3 4 5 6 7
```

Note that the function to print the board is also given to you in the git file

2. Ask the first player to play and print the updated board.

If the player is a human it should print the following message:

Player {PLAYER NUMBER} (X) turn.

Enter column (1-{NUMBER OF COLUMNS}):

And let the player choose a column to drop the token at.

If the player is a computer it should print the following messages:

Player {PLAYER NUMBER} (X) turn.

Computer chose column {CHOSEN COLUMN}

with the chosen column of the computer.

and then print the updated board. for example:

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|X|.|.|.|

 1 2 3 4 5 6 7

3. Ask the second player to play and print the updated board.

4. Continue in this manner until the end of the game.

For example:

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

 1 2 3 4 5 6 7

Player 1 (X) turn.

Enter column (1-7): 4

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|X|.|.|.|

 1 2 3 4 5 6 7

Player 2 (O) turn.

Computer chose column 4

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|O|.|.|.|

|.|.|.|X|.|.|.|

 1 2 3 4 5 6 7

Player 1 (X) turn.

Enter column (1-7): 3

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|O|.|.|.|

|.|.|X|X|.|.|.|

 1 2 3 4 5 6 7

Player 2 (O) turn.

Computer chose column 5

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|.|.|.|.|

|.|.|.|O|.|.|.|

|.|.|X|X|O|.|.|

 1 2 3 4 5 6 7

Player 1 (X) turn.

Enter column (1-7):

In this Example, we chose the first player to be a human - “h”, and the second player to be a computer - “c”.

Then, we printed the empty board.

At the start - we waited for input [a number for the column], and printed the board after his choice.

Then, because it was the computer turn, we printed the board after the computer choice immediately [there is no input waiting time].

And so on.

The logic above is also relevant for 2 human players [just do the human logic in each turn], and 2 computer players [just print their choices until one wins].

At the end of the game, print an appropriate victory message after printing the winning board:

|.|.|O|O|X|X|.|

|.|.|X|X|X|O|.|

|.|O|O|O|O|X|.|

|.|X|X|X|O|O|X|

|.|X|O|O|X|X|O|

|O|O|O|X|X|X|O|

 1 2 3 4 5 6 7

Player 2 (O) wins!

If no free columns remain and no player has created a sequence of 4 tokens - the game ends in a tie.

print the following message:

Board full and no winner. It's a tie!

Victory detection should be performed on every move in four directions: vertical, horizontal and diagonals.

Behavior of the Human player

This player may choose any column. If the chosen column is illegal, a message will be displayed and the player will be asked to choose again.

As follows:

Enter column (1-7): 8

Invalid column. Choose between 1 and 7.

Player 1 (X) turn.

Enter column (1-7): 4

Column 4 is full. Choose another column.

Enter column (1-7): a

Invalid input. Enter a number.

Enter column (1-7):

d

Invalid input. Enter a number.

Enter column (1-7): -1

Invalid column. Choose between 1 and 7.

Enter column (1-7):

Please notice that the behavior of invalid input is:

If it’s a number that is not in range of the valid columns - we will output: “Invalid column. Choose between 1 and b.” When b is the number of columns we have.

If it’s a number in the valid range, but the column is already full, we will print “Column x is full. Choose another column.” When x is the number the user tried to enter.

If it’s not a number, and not white space character, we will output: “Invalid input. Enter a number.”

If it’s a white-space character, we will ignore it.

Behavior of the computer player

This player will choose a column according to a defined priority order. If several columns share the same priority level they should be chosen according to the ordering rule described below.

Priority order

1. Winning move - if it is possible to win on the next move - choose the column that produces the win.

2. Blocking the opponent - if the opponent can win on their next move choose the column that prevents this.

3. Creating a sequence of three - if it is possible to create a sequence of three tokens do so.

4. Blocking the opponent’s sequence of three.

5. Choosing according to the arbitrary ordering rule.

Programming tip: each type of player (human or computer) should return only the chosen column number. The game engine is responsible for checking legality, performing the move and updating the board.

Ordering rule for columns

When several options have the same priority level, choose a column according to the following rules:

1. Prefer the column whose distance from the center column is minimal.

2. If the distance is equal, choose the left column among the two.

For example:

If there are 7 columns the ordering is:

4 -> 3 -> 5 -> 2 -> 6 -> 1 -> 7

If there are 6 columns the ordering is:

3 -> 4 -> 2 -> 5 -> 1 -> 6

Full running Example. 4 X 4 board - two computers

```
Connect Four (4 rows x 4 cols)

Choose type for player 1: h - human, c - computer: c

Choose type for player 2: h - human, c - computer: c

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

 1 2 3 4

Player 1 (X) turn.

Computer chose column 2

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|.|

 1 2 3 4
```

Player 2 (O) turn.

Computer chose column 2

|.|.|.|.|

|.|.|.|.|

|.|O|.|.|

|.|X|.|.|

 1 2 3 4

Player 1 (X) turn.

Computer chose column 2

|.|.|.|.|

|.|X|.|.|

|.|O|.|.|

|.|X|.|.|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 2

|.|O|.|.|

|.|X|.|.|

|.|O|.|.|

|.|X|.|.|

 1 2 3 4

Player 1 (X) turn.

Computer chose column 3

|.|O|.|.|

|.|X|.|.|

|.|O|.|.|

|.|X|X|.|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 1

|.|O|.|.|

|.|X|.|.|

|.|O|.|.|

|O|X|X|.|

 1 2 3 4

Player 1 (X) turn.

Computer chose column 4

|.|O|.|.|

|.|X|.|.|

|.|O|.|.|

|O|X|X|X|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 3

|.|O|.|.|

|.|X|.|.|

|.|O|O|.|

|O|X|X|X|

 1 2 3 4

Player 1 (X) turn.

Computer chose column 3

|.|O|.|.|

|.|X|X|.|

|.|O|O|.|

|O|X|X|X|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 1

|.|O|.|.|

|.|X|X|.|

|O|O|O|.|

|O|X|X|X|

 1 2 3 4

Player 1 (X) turn.

Computer chose column 4

|.|O|.|.|

|.|X|X|.|

|O|O|O|X|

|O|X|X|X|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 1

|.|O|.|.|

|O|X|X|.|

|O|O|O|X|

|O|X|X|X|

 1 2 3 4

Player 1 (X) turn.

Computer chose column 1

|X|O|.|.|

|O|X|X|.|

|O|O|O|X|

|O|X|X|X|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 4

|X|O|.|.|

|O|X|X|O|

|O|O|O|X|

|O|X|X|X|

 1 2 3 4

Player 1 (X) turn.

Computer chose column 3

|X|O|X|.|

|O|X|X|O|

|O|O|O|X|

|O|X|X|X|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 4

|X|O|X|O|

|O|X|X|O|

|O|O|O|X|

|O|X|X|X|

 1 2 3 4

Board full and no winner. It's a tie!

Full running Example. 10 X 4 board - human and computer

Connect Four (10 rows x 4 cols)

Choose type for player 1: h - human, c - computer: h

Choose type for player 2: h - human, c - computer: c

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

 1 2 3 4

Player 1 (X) turn.

Enter column (1-4): 2

|.|.|.|.|

|.|.|.|.|

```
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|X|.|.|
 1 2 3 4
```

Player 2 (O) turn.

Computer chose column 2

```
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|O|.|.|
|.|X|.|.|
 1 2 3 4
```

Player 1 (X) turn.

Enter column (1-4): 3

```
|.|.|.|.|
```

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|O|.|.|

|.|X|X|.|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 1

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|O|.|.|

|O|X|X|.|

 1 2 3 4

Player 1 (X) turn.

Enter column (1-4): 3

```
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|O|X|.|
|O|X|X|.|
 1 2 3 4
```

Player 2 (O) turn.

Computer chose column 3

```
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|.|.|
|.|.|O|.|
|.|O|X|.|
|O|X|X|.|
 1 2 3 4
```

Player 1 (X) turn.

Enter column (1-4): 4

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|.|O|X|.|

|O|X|X|X|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 2

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|O|O|.|

|.|O|X|.|

|O|X|X|X|

 1 2 3 4

```
Player 1 (X) turn.

Enter column (1-4): 4

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|O|O|.|

|.|O|X|X|

|O|X|X|X|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 2

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|O|.|.|

|.|O|O|.|

|.|O|X|X|

|O|X|X|X|
```

```
 1 2 3 4

Player 1 (X) turn.

Enter column (1-4): 2

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|.|

|.|O|.|.|

|.|O|O|.|

|.|O|X|X|

|O|X|X|X|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 4

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|.|

|.|O|.|.|

|.|O|O|O|

|.|O|X|X|
```

```
|O|X|X|X|

 1 2 3 4

Player 1 (X) turn.

Enter column (1-4): 1

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|.|

|.|O|.|.|

|.|O|O|O|

|X|O|X|X|

|O|X|X|X|

 1 2 3 4

Player 2 (O) turn.

Computer chose column 1

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|.|

|.|O|.|.|

|O|O|O|O|
```

|X|O|X|X|

|O|X|X|X|

 1 2 3 4

Player 2 (O) wins!

Full running Example. 5 X 5 board - human and human

Connect Four (5 rows x 5 cols)

Choose type for player 1: h - human, c - computer: h

Choose type for player 2: h - human, c - computer: h

|.|.|.|.|.|

|.|.|.|.|.|

|.|.|.|.|.|

|.|.|.|.|.|

|.|.|.|.|.|

 1 2 3 4 5

Player 1 (X) turn.

Enter column (1-5): 0

Invalid column. Choose between 1 and 5.

Enter column (1-5): a

Invalid input. Enter a number.

Enter column (1-5):

7

Invalid column. Choose between 1 and 5.

Enter column (1-5): 3

|.|.|.|.|.|

|.|.|.|.|.|

```
|.|.|.|.|.|
|.|.|.|.|.|
|.|.|X|.|.|
 1 2 3 4 5
Player 2 (O) turn.
Enter column (1-5): 1
|.|.|.|.|.|
|.|.|.|.|.|
|.|.|.|.|.|
|.|.|.|.|.|
|O|.|X|.|.|
 1 2 3 4 5
Player 1 (X) turn.
Enter column (1-5): 3
|.|.|.|.|.|
|.|.|.|.|.|
|.|.|.|.|.|
|.|.|X|.|.|
|O|.|X|.|.|
 1 2 3 4 5
Player 2 (O) turn.
Enter column (1-5): 2
|.|.|.|.|.|
|.|.|.|.|.|
|.|.|.|.|.|
```

|.|.|X|.|.|

|O|O|X|.|.|

 1 2 3 4 5

Player 1 (X) turn.

Enter column (1-5): 3

|.|.|.|.|.|

|.|.|.|.|.|

|.|.|X|.|.|

|.|.|X|.|.|

|O|O|X|.|.|

 1 2 3 4 5

Player 2 (O) turn.

Enter column (1-5): 4

|.|.|.|.|.|

|.|.|.|.|.|

|.|.|X|.|.|

|.|.|X|.|.|

|O|O|X|O|.|

 1 2 3 4 5

Player 1 (X) turn.

Enter column (1-5): 3

|.|.|.|.|.|

|.|.|X|.|.|

|.|.|X|.|.|

|.|.|X|.|.|

|O|O|X|O|.|

 1 2 3 4 5

Player 1 (X) wins!

Good Luck!

אני מצרף לך את הקוד התקין שהיה בעבור מטלה 3 לצורך השראה והבנת המערכת

פתור את משימה 7 במלואה ושים לב לשינוי המהותי בה

Exercise 7 - Upgraded Connect- 4 (Connect-N)

We all remember the good old days when the sun was shining, and trees' roots weren't upside down. This brings us back to simpler times when our great concern was ex3.

Your task is simple - recreate ex3, but in python using any AI tool you want (prompts you used will be submitted in a separate pdf file, did I mention that?), with a few key changes:

The function for printing the board is not given.

The board's dimensions are received from the user as inputs and need to be validated, and the game's goal will be adjusted accordingly by the logic explained later.

N-Connect Logic:

Since we want to make our board's size adjustable, we are going to change the length of the sequence we need to mark.

Anything less than 2 or greater than 100 (101 and above, rows or columns) - invalid.

If 2 is chosen for either rows or columns, sequence is 2 (very short game, but that is the user's responsibility)

If either is 3 - game transforms to tic-tac-toe and the board is 3X3 (Ix Igul), two human players only. You can assume no one is going to mark a taken cell.

4 <= rows or columns <= 5 sequence is 3

6 <= rows or columns <= 10 sequence is 4

11 and above - sequence is 5

Tic-Tac-Toe:

Users will need to enter 1 number for the cell chosen. Cells are numbered in this format:

|1|2|3|

|4|5|6|

|7|8|9|

Columns will not be numbered

Good Luck!

All other examples are the same as ex3

פשוט עבור פייתון מצורף CODING STYLEבנוסף הקוד אמור להיות עדיין לפי חוקי ה
פשוט תמיר את ההבנה הזאת לפייתון C הקובץ עם ההוראות עבור

Step 2: Debugging Tic-Tac-Toe Logic

תקן באגים:

Choose type for player 1: h - human, c - computer: c

Choose type for player 2: h - human, c - computer: c

|1|2|3|

|4|5|6|

|7|8|9|

Player 1 (X) turn.

Enter cell (1-9): 8

|1|2|3|

|4|5|6|

|7|X|9|

Player 2 (O) turn.

Enter cell (1-9): 9

|1|2|3|

|4|5|6|

|7|X|O|

Player 1 (X) turn.

Enter cell (1-9): 0

Invalid cell. Choose between 1 and 9.

Enter cell (1-9): 8

Cell is full. Choose another cell.

Enter cell (1-9): 6

|1|2|3|

|4|5|X|

|7|X|O|

Player 2 (O) turn.

Enter cell (1-9): 5

|1|2|3|

|4|O|X|

|7|X|O|

Player 1 (X) turn.

Enter cell (1-9): 4

|1|2|3|

|X|O|X|

|7|X|O|

Player 2 (O) turn.

Enter cell (1-9): 6

Cell is full. Choose another cell.

Enter cell (1-9): 5

Cell is full. Choose another cell.

Enter cell (1-9): 6

Cell is full. Choose another cell.

Enter cell (1-9): 7

|1|2|3|

|X|O|X|

|O|X|O|

Player 1 (X) turn.

Enter cell (1-9): 8

Cell is full. Choose another cell.

Enter cell (1-9): 5

Cell is full. Choose another cell.

Enter cell (1-9): 4

Cell is full. Choose another cell.

Enter cell (1-9): 1

|X|2|3|

|X|O|X|

|O|X|O|

Player 2 (O) turn.

Enter cell (1-9): 2

|X|O|3|

|X|O|X|

|O|X|O|

Player 1 (X) turn.

Enter cell (1-9): 3

|X|O|X|

|X|O|X|

|O|X|O|

Board full and no winner. It's a tie!

למרות שנבחרו משחק מחשב מול מחשב הוא מבקש מהמשתמש

Step 3: Matching Output Format

ככה זה אמור להראות לדוגמה:

Enter number of rows

3

Enter number of columns

9

Tic Tac Toe (Human vs Human)

|.|.|.|

|.|.|.|

|.|.|.|

Enter position (1-9): 1

|X|.|.|

|.|.|.|

|.|.|.|

Enter position (1-9): 3

|X|.|O|

|.|.|.|

|.|.|.|

Enter position (1-9): 5

|X|.|O|

|.|X|.|

|.|.|.|

Enter position (1-9):

ולא ככה:

Enter number of rows: 3

Enter number of columns: 9

Connect Four (3 rows x 3 cols)

|1|2|3|

|4|5|6|

|7|8|9|

Player 1 (X) turn.

Enter cell (1-9): 3

|1|2|X|

|4|5|6|

|7|8|9|

Player 2 (O) turn.

Enter cell (1-9): 9

|1|2|X|

|4|5|6|

|7|8|O|

Player 1 (X) turn.

Enter cell (1-9): 6

|1|2|X|

|4|5|X|

|7|8|O|

Player 2 (O) turn.

Enter cell (1-9): 1

|O|2|X|

|4|5|X|

|7|8|O|

Player 1 (X) turn.

Enter cell (1-9): 2

|O|X|X|

|4|5|X|

|7|8|O|

Player 2 (O) turn.

Enter cell (1-9): 3

Cell is full. Choose another cell.

Enter cell (1-9): 4

|O|X|X|

|O|5|X|

|7|8|O|

וגם ככה:

Enter number of rows

4

Enter number of columns

4

Connect Four - Or More [Or Less] (4 rows x 4 cols, connect 3)

Choose type for player 1: h - human, r - random/simple computer, s - strategic computer:

ולא ככה:

Enter number of rows: 4

Enter number of columns: 4

Connect Four (4 rows x 4 cols)

Choose type for player 1: h - human, c - computer: c

Choose type for player 2: h - human, c - computer: c

וגם כך וכו:

Enter number of rows

2

Enter number of columns

8

Connect Four - Or More [Or Less] (2 rows x 8 cols, connect 2)

Choose type for player 1: h - human, r - random/simple computer, s - strategic computer:

Step 4: Final Polish (Style & Input)

it should be in the same line:

Choose type for player 1: h - human, r - random/simple computer, s - strategic computer: r

fix it

also fix the use of magic numbers

Step 5: Final Adjustment (Forum Clarification):

קיבלתי את ההבהרה הבאה תקן את הקוד בהתאם:

שבוע טוב לכולם

עקב כל מיני בלבולים שנוצרו עם ההוראות לתרגיל, להלן כמה הבהרות:

מה שעשיתם בתרגיל 3 - אותו דבר, רק בפייתון

לגבי הכנסת 2 ו3 - טעות שלי, צריך שיהיה איקס עיגול

בנוסף, הטסטים שנריץ לבדיקות יהיו אותם הטסטים שיחזרו בבדיקה האוטומטית.

רק אל תשכחו לצרף את קובץ הפרומפטים

בתגובה ל: שלום אדורם

תשובה ל: הבהרה

על ידי דניאל בלנקי בתאריך יום ראשון, 25 ינואר 2026, 15:44

מה אנחנו צריכים לעשות ב-random/simple computer וב-strategic computer? זה לא היה לנו בתרגיל 3, אבל מופיע בקובץ לדוגמא. אז צריך להתעלם מזה ופשוט לעשות כמו בתרגיל 3?

בתגובה ל: דניאל בלנקי

תשובה ל: הבהרה

על ידי שלום אדורם בתאריך יום ראשון, 25 ינואר 2026, 16:39

נכון

Step 6:

יש שגיאות בפלט ובלוגיקה תן לפי הטעויות המופיעות כאן:

Your grade for assignment "ex7" is "50".

"ex7" was checked with 2 inputs:

(1) 0 line errors, , 50 points out of 50.

Input

3

3

1

5

9

2

8

7

4

3

Correct Output

Enter number of rows

Enter number of columns

Tic Tac Toe (Human vs Human)

|.|.|.|

|.|.|.|

|.|.|.|

Enter position (1-9):

|X|.|.|

|.|.|.|

|.|.|.|

Enter position (1-9):

|X|.|.|

|.|O|.|

|.|.|.|

Enter position (1-9):

|X|.|.|

|.|O|.|

|.|.|X|

Enter position (1-9):

|X|O|.|

|.|O|.|

|.|.|X|

Enter position (1-9):

|X|O|.|

|.|O|.|

|.|X|X|

Enter position (1-9):

|X|O|.|

|.|O|.|

|O|X|X|

Enter position (1-9):

|X|O|.|

|X|O|.|

|O|X|X|

Enter position (1-9):

|X|O|O|

|X|O|.|

|O|X|X|

Player 2 (O) wins!

Your Output

Enter number of rows

Enter number of columns

Tic Tac Toe (Human vs Human)

|.|.|.|

|.|.|.|

|.|.|.|

Enter position (1-9):

|X|.|.|

|.|.|.|

|.|.|.|

Enter position (1-9):

|X|.|.|

|.|O|.|

|.|.|.|

Enter position (1-9):

|X|.|.|

|.|O|.|

|.|.|X|

Enter position (1-9):

|X|O|.|

|.|O|.|

|.|.|X|

Enter position (1-9):

|X|O|.|

|.|O|.|

|.|X|X|

Enter position (1-9):

|X|O|.|

|.|O|.|

|O|X|X|

Enter position (1-9):

|X|O|.|

|X|O|.|

|O|X|X|

Enter position (1-9):

|X|O|O|

|X|O|.|

|O|X|X|

Player 2 (O) wins!

(2) 13 line errors, , 0 points out of 50.

Input

4

4

h

h

2

4

3

3

1

Correct Output

Enter number of rows

Enter number of columns

Connect Four - Or More [Or Less] (4 rows x 4 cols, connect 3)

Choose type for player 1: h - human, r - random/simple computer, s - strategic computer: Choose type for player 2: h - human, r - random/simple computer, s - strategic computer:

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|.|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|X|O|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|.|X|X|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|X|X|X|O|

1 2 3 4

Player 1 (X) wins!

Your Output

Enter number of rows

Enter number of columns

Connect Four (4 rows x 4 cols)

Choose type for player 1: h - human, c - computer: Choose type for player 2: h - human, c - computer: |.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4): |.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|.|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4): |.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4): |.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|X|O|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4): |.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|.|X|X|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4): |.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|X|X|X|O|

1 2 3 4

Player 1 (X) wins!

Step 7: Final Fix for Auto grader Newline Issue:
fix the code according to this:

Your grade for assignment "ex7" is "0".

"ex7" was checked with 2 inputs:

(1) 16 line errors, , 0 points out of 50.

Input

3

3

1

5

9

2

8

7

4

3

Correct Output

Enter number of rows

Enter number of columns

Tic Tac Toe (Human vs Human)

|.|.|.|

|.|.|.|

|.|.|.|

Enter position (1-9):

|X|.|.|

|.|.|.|

|.|.|.|

Enter position (1-9):

|X|.|.|

|.|O|.|

|.|.|.|

Enter position (1-9):

|X|.|.|

|.|O|.|

|.|.|X|

Enter position (1-9):

|X|O|.|

|.|O|.|

|.|.|X|

Enter position (1-9):

|X|O|.|

|.|O|.|

|.|X|X|

Enter position (1-9):

|X|O|.|

|.|O|.|

|O|X|X|

Enter position (1-9):

|X|O|.|

|X|O|.|

|O|X|X|

Enter position (1-9):

|X|O|O|

|X|O|.|

|O|X|X|

Player 2 (O) wins!

Your Output

Enter number of rows

Enter number of columns

Tic Tac Toe (Human vs Human)

|.|.|.|

|.|.|.|

|.|.|.|

Enter position (1-9): |X|.|.|

|.|.|.|

|.|.|.|

Enter position (1-9): |X|.|.|

|.|O|.|

|.|.|.|

Enter position (1-9): |X|.|.|

|.|O|.|

|.|.|X|

Enter position (1-9): |X|O|.|

|.|O|.|

|.|.|X|

Enter position (1-9): |X|O|.|

|.|O|.|

|.|X|X|

Enter position (1-9): |X|O|.|

|.|O|.|

|O|X|X|

Enter position (1-9): |X|O|.|

|X|O|.|

|O|X|X|

Enter position (1-9): |X|O|O|

|X|O|.|

|O|X|X|

Player 2 (O) wins!

(2) 12 line errors, , 0 points out of 50.

Input

4

4

h

h

2

4

3

3

1

Correct Output

Enter number of rows

Enter number of columns

Connect Four - Or More [Or Less] (4 rows x 4 cols, connect 3)

Choose type for player 1: h - human, r - random/simple computer, s - strategic computer: Choose type for player 2: h - human, r - random/simple computer, s - strategic computer:

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|.|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|X|O|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|.|X|X|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|X|X|X|O|

1 2 3 4

Player 1 (X) wins!

Your Output

Enter number of rows

Enter number of columns

Connect Four - Or More [Or Less] (4 rows x 4 cols, connect 3)

Choose type for player 1: h - human, r - random/simple computer, s - strategic computer: Choose type for player 2: h - human, r - random/simple computer, s - strategic computer: |.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4): |.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|.|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4): |.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4): |.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|X|O|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4): |.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|.|X|X|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4): |.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|X|X|X|O|

1 2 3 4

Player 1 (X) wins!

Step 8: Final Separation of Board and Prompts:

I got this mistake fix it:

(2) 2 line errors, , 0 points out of 50.

Input

4

4

h

h

2

4

3

3

1

Correct Output

Enter number of rows

Enter number of columns

Connect Four - Or More [Or Less] (4 rows x 4 cols, connect 3)

Choose type for player 1: h - human, r - random/simple computer, s - strategic computer: Choose type for player 2: h - human, r - random/simple computer, s - strategic computer:

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|.|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|X|O|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|.|X|X|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|X|X|X|O|

1 2 3 4

Player 1 (X) wins!

Your Output

Enter number of rows

Enter number of columns

Connect Four - Or More [Or Less] (4 rows x 4 cols, connect 3)

Choose type for player 1: h - human, r - random/simple computer, s - strategic computer: Choose type for player 2: h - human, r - random/simple computer, s - strategic computer: |.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|.|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|.|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|.|.|

|.|X|X|O|

1 2 3 4

Player 2 (O) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|.|X|X|O|

1 2 3 4

Player 1 (X) turn.

Enter column (1-4):

|.|.|.|.|

|.|.|.|.|

|.|.|O|.|

|X|X|X|O|

1 2 3 4

Player 1 (X) wins!